Revanth Umakanth

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Summary

Mechanical Engineering student at Vellore Institute of Technology with hands-on experience in manufacturing process analysis, production line optimization, and Continuous Improvement using Lean Manufacturing and Six Sigma methodologies. Skilled in time-and-motion study, cycle-time analysis, and manufacturing data analysis to identify production bottlenecks, evaluate process-improvement opportunities, and support engineering problem-solving in a production environment. Additional experience in mechatronics, embedded systems, and automation.

Core Competencies: Lean Manufacturing | Six Sigma | Continuous Improvement (CI) | Process Optimization | Time and Motion Study | Cycle-Time Analysis | Manufacturing Data Analysis | Production Bottleneck Identification | Automation | Mechatronics | Embedded Systems

Education

Vellore Institute of Technology (VIT) — Vellore, Tamil Nadu
B.Tech, Mechanical Engineering | CGPA: 8.04/10 — *2023 - 2027*

Experience

Industrial Engineering Intern — *Jul 2026*
Amara Raja Energy & Mobility Ltd. (AMARON Batteries) — *Tirupati, India*

- *Interned in the CI and Lean department; assessed manual operations for automation potential based on productivity, consistency, and operator workload.*
- *Studied a hydraulic pick-and-place system for heavy industrial batteries and plotted routes for a planned tow-truck battery-transport system.*
- *Conducted time-and-motion and cycle-time analysis (VideoTimerPro), applying Lean and Six Sigma methodologies to identify production bottlenecks.*
- *Proposed process improvements projected to reduce fixed delay time by 16.56 percent and improve full insertion process efficiency by 7.14 percent.*

Engineering Intern — *Jun 2025 - Jul 2025*
Edomotics Smart Systems Pvt. Ltd. — *Hyderabad, India*

- *Worked on mechatronics and home automation systems involving sensor and actuator integration and Control4 platforms, supporting embedded systems development and hardware-software interfacing for automated control applications.*

Projects

AI-Assisted Route Optimization for Heavy-Duty Trucks | *Python, Machine Learning, Random Forest, XGBoost, Gemini API, OSMnx, Geospatial Analysis*

- *Built a machine learning-based route optimization system for heavy-duty trucks, applying data-driven decision-making and engineering problem-solving to reduce fuel consumption, braking events, and carbon emissions on long-haul routes.*
- *Engineered features from GPS and road-network data using OSMnx, and trained and validated Random Forest and XGBoost regression models to evaluate route characteristics and predict optimal routes.*
- *Integrated the Gemini API to generate natural-language route summaries and driver-facing recommendations for AI-assisted route analysis.*

Development of an Edge-AI-Based Intelligent Pick-and-Place Robotic System with Material-Based Drop Height Control | *Python, Machine Learning, Edge AI, Image Processing, Raspberry Pi, Dobot Magician Robotic Arm (Capstone Project)*

- *Developed an Edge AI-based pick-and-place robotic system that classifies an object's material (metal, glass, wood, or plastic) and selects a suitable drop height for safe, efficient handling, using a Dobot Magician 4-DOF robotic arm controlled via Raspberry Pi.*
- *Built and tested a basic pick-and-place operation, integrated the Raspberry Pi with the Dobot, and developed robot-control commands through the Raspberry Pi terminal.*
- *Collected and prepared an image dataset across the four material categories, and trained a machine learning model for image-based material classification to enable material-based drop-height selection.*

Technical Skills

Manufacturing & Continuous Improvement: *Lean Manufacturing, Continuous Improvement (CI), Process Optimization, Six Sigma, 5S, Time and Motion Study, Cycle Time Analysis*
Data & Analysis: *Excel, Python, MATLAB, Data Analysis*
Automation & Mechatronics: *Automation, Sensors and Actuators, Arduino, PID Control, Embedded Systems, Mechatronics, Control4*
Design & CAD: *SolidWorks, CAD Modeling, 3D Printing, Rapid Prototyping*
Machine Learning: *Random Forest, XGBoost, Scikit-learn, Gemini API*
Simulation & Analysis: *ANSYS APDL (FEA, meshing, structural analysis)*

Certifications

- ***Certificate in Collaborative Robotics in Manufacturing*** - *L&T (Larsen & Toubro)* — *22 May 2025*
- *Completed training in industrial robotics, collaborative robots (cobots), robot programming, automation integration, end-effectors, and industrial safety.*